

Harsath Kaliappa Gounder Thangavel

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Education

York University

Bachelor of Science in Computer Science (Honours)

January 2020 – Present

Toronto, Canada

- Data structures
- Database systems
- Computer networking
- Compiler design
- Computer architecture
- Operating systems
- Data mining
- Linguistics

Work Experience

CogniPet

Computer vision engineer

April 2019 – November 2019

Zürich, Switzerland

- Trained and deployed a deep learning model for recognizing over 120 breeds of dogs using the Inception V3 neural network and transfer learning technique.
- Implemented a REST API to run model inference and integrated it with CogniPet's Android and iOS platforms using Amazon Web Services (AWS).

Projects

Prism Interpreter | *Interpreter for the Prism programming language*

- Implemented an interpreter for a language that supports recursion, arrays, classes (user-defined datatypes), functions, nested *for* and *while* loops, built-in functions, objects, and various built-in data types including integers, strings, and booleans.
- Interpreter and code coverage engine built using ANTLR4 parser generator in Java.

Blueth (libblueth) | *Modern C++17 Linux networking framework*

- Networking framework written in modern C++17 for Linux and provides a set of common client-server abstractions. Enables development of high-performance, concurrent, and event-driven networking services in a modular and composable manner.
- Includes event-loop abstraction, HTTP parsers, proxies, thread-pool executors, caches, common codecs, SIMD wrappers, and networking utilities.

Cap-Containers | *C header-only containers library*

- Highly-generic, self-contained, header-only containers written in C.
- Implements sets, maps, vectors, hash-tables (open and closed addressing), queues (circular, dynamic, fixed, and priority queues), stack, lists, and arena memory allocator.
- Includes thread-safe variants with cross-platform support (Windows and UNIX).
- Tested using CI/CD pipelines and extensively documented with Doxygen.

Poc-HTTP-Parser | *C header-only high-performance HTTP parsers*

- SIMD vectorized (SSE and AVX2), RFC-compliant, HTTP 1.x parsers and state-machines.
- Tested against common vulnerabilities such as HTTP request smuggling.
- Benchmarked using Google's microbenchmark library and compared its performance against Fastly's parsers.

Dog breed recognition using Inception V3 neural network | *Deep learning model*

- Trained a deep learning model to recognize over 120 breeds of dogs using Google Colab with GPU acceleration.
- Wrote a REST API for running inference over the trained machine learning model for CogniPet's minimum viable product (MVP) Android and iOS apps.

Technical Skills

Languages: Python, modern C++, C, Java, SQL, PHP, $\text{\LaTeX}$, HTML/CSS, Bash, Perl, Golang
Technologies: Linux, Docker, Git, LXC, MySQL, Postfix, Dovecot, CMake
Developer tools: Vim, GDB, VSCode, Amazon Web Services

Hobbies

Wikipedia

Editor

July 2021 – Present

- Made over 20,000 edits to mainspace Wikipedia and have significantly contributed to articles in various domains, including computer networking, data structures, network security, neuroscience, pharmacology, and history.